🌾 AgroSetu — Product Roadmap & Architecture

> Version 3.0 · March 2026 · Umiya AgroSetu Digital LLP
> Replaces: AgroSetu_Product_Roadmap_v2.docx

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1. What Has Changed Since v2.0

Area	v2.0 Status	v3.0 Reality
WhatsApp (Twilio)	Sandbox only, "join keyword" friction	✅ **Twilio testing complete**
Meta WhatsApp API	Blocked, community case raised	✅ **Meta developer account now active**
Broker login	Not built	✅ **Built — full broker portal**
Brokerage ledger	Highest priority, not started	✅ **Built — trade entry, settlement, invoices**
Railway deployment	Pending	⏳ Waiting on LLP name approval
MVP phase naming	Phase 0 / Phase 1	Renamed to MVP 1.x milestones

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2. MVP Milestone Tracker

✅ MVP 1.0 — Foundation (Oct–Dec 2025)

> Admin rate entry + public website + WhatsApp broadcast

- Daily rate entry admin panel (`price-entry.html`, localhost:5000)
- Public rates page — `umiyaagrosetu.com` live on Netlify
- WhatsApp broadcast via Twilio (Gujarati-formatted message)
- Admin login protection (session-based)
- TradingView MCX Cotton + NCDEX Castor Seed widgets
- Visitor IP logging

✅ MVP 1.5 — Broker Portal (Jan–Mar 2026)

> Full broker login + brokerage ledger + settlement invoices

Broker Authentication

- `broker-login.html` — session-based login per broker
- `brokerAuth.js` middleware — route protection
- Broker "me" API — `GET /api/broker/me`
- Logout endpoint

Trade Management

- `broker-entry.html` — trade entry with factory + wholesaler + commodity dropdowns
- `broker-trades.html` — trade history with date filters, search, edit/delete
- Trade API — full CRUD (`GET/POST/PUT/DELETE /api/broker/trades`)

Settlement & Invoicing

- `broker-report.html` — settlement overview + invoice generator
- Grand totals dashboard (factory due / wholesaler due / outstanding)
- Per-party settlement cards with running balance
- Mark Settlement modal — records payment with date and remarks
- Settlement history toggle per party
- **Invoice generator** — `AGS-YYYYFYY-NNN` numbered PDF invoices
- PDF download via html2canvas + jsPDF (multi-page, A4)
- WhatsApp + PDF flow — downloads PDF, opens WhatsApp web pre-filled

Admin — Master Data

- `admin.html` — full CRUD for Brokers, Factories, Wholesalers, Commodities
- `masterDataRoutes.js` — consolidated route (replaced `adminBrokerRoutes.js`)
- Assign drawer — broker ↔ factory and broker ↔ wholesaler assignments
- City field across all entities
- `apiFetch()` helper — auto-redirects to login on 401

Database additions (broker schema)

- `brokers` table — login credentials, city, status
- `trades` table — full trade records
- `brokerage_settlements` table — settlement records per party per FY
- `broker_factories`, `broker_wholesalers` — assignment junction tables

- FY-aware queries (April–March cycle)

🔄 MVP 1.6 — Public Daily Rates Page v2 (Next — Apr 2026)

> Replace hardcoded/API-driven public page with a simple CSV-powered static page

Goal: Broker (or admin) drops a CSV file → public page auto-renders today's rates. Zero backend dependency for the public-facing page.

Why CSV?

- Backend not yet deployed publicly (waiting on LLP + AWS)
- Netlify can serve a static page that reads a CSV from the same repo
- Father can update rates without touching code — just update one file
- Fallback when backend is down

Single Page Design

```
umiyaagrosetu.com/rates  (or index page)
  └── rates.csv          (committed to GitHub / Netlify)
      factory, commodity, price_per_bag, unit, date
      "Shree Bhagvati Mill", "Jaadu Bhusu 39kg", 901, bag, 2026-03-12
      ...
  └── index.html
      Fetches rates.csv at load time (Papa Parse or native fetch)
      Renders rate cards grouped by factory
      Shows "Last updated: 12 Mar 2026"
      Gujarati + English commodity names
      Mobile responsive
      SEO meta tags + JSON-LD structured data
```

Build tasks

- [ ] Design `rates.csv` schema (factory, commodity, price, unit, date, remark_guj)
- [ ] Build `index.html` — CSV fetch + render with Papa Parse
- [ ] Add "Last Updated" timestamp from CSV `date` column
- [ ] Gujarati commodity name column in CSV → display Gujarati on page
- [ ] SEO: `<title>`, `<meta description>`, Open Graph, JSON-LD PriceSpecification
- [ ] Deploy to Netlify — `git push` = rates go live in 60 seconds
- [ ] Update admin panel — "Export today's rates as CSV" button (one click → download `rates.csv`)

Future bridge: Once backend is on AWS, the page can switch from CSV fetch to API fetch — zero design change needed.

⏳ MVP 1.7 — WhatsApp via Meta API (Apr–May 2026)

> Replace Twilio with Meta WhatsApp Cloud API (developer account now active)

Status: Meta developer account active ✅ · Twilio testing complete ✅

Migration plan

- [ ] Create WhatsApp Business message template in Meta dashboard (Gujarati daily rates format)
- [ ] Get template approved (typically 24–48 hrs)
- [ ] Update `whatsappService.js` — swap Twilio client for Meta Cloud API calls
- [ ] Update `whatsappConfig.js` — `WHATSAPP_TOKEN` + `WHATSAPP_PHONE_NUMBER_ID` from `.env`
- [ ] Test broadcast end-to-end with father's number
- [ ] Add subscriber opt-in flow (Meta requires explicit opt-in)
- [ ] Keep Twilio as fallback until Meta fully stable

Why this matters: Meta API has no "join keyword" friction — subscribers receive messages directly once opted in. Unlocks proper WhatsApp Business broadcast at scale.

⏳ MVP 2.0 — Cloud Deployment (May–Jun 2026)

> Backend live on AWS, real data flowing to public website

Blocked on: LLP name approval (Umiya AgroSetu Digital LLP)

- [ ] AWS account setup (EC2 + RDS, ap-south-1 Mumbai)
- [ ] Migrate PostgreSQL from local → RDS
- [ ] Deploy Node.js backend → EC2 (or Elastic Beanstalk)
- [ ] Configure CORS — Netlify origin whitelisted
- [ ] Update `CONFIG.siteUrl` → `umiyaagrosetu.com`
- [ ] SSL on backend domain (`api.umiyaagrosetu.com`)
- [ ] Switch public rates page from CSV → `GET /api/rates/today`
- [ ] Set up `umiyaagrosetu.in` as Netlify domain alias
- [ ] Windows Task Scheduler → daily PostgreSQL backup script

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⏳ MVP 2.5 — Multi-Broker Platform (Jul–Sep 2026)

- Broker self-registration (admin approves)

- Each broker has isolated factory/wholesaler network
- Isolated brokerage data per broker (row-level security)
- Subscription billing via Razorpay (₹999/mo broker plan)
- Price history charts (30-day trends)
- Admin super-dashboard across all brokers

⏳ Phase 3 — Factory & Wholesaler Portals (Sep–Dec 2026)

- Factory self-signup → rate publishing portal
- Factory brokerage dues dashboard (self-service)
- Wholesaler portal — browse rates, place orders, view dues
- Progressive Web App (PWA) — mobile first
- Order history and digital payment records

⏳ Phase 4 — Marketplace & Scale (Jan 2027+)

- Farmer price discovery and direct buying
- Transaction commission model (₹1–2/bag on digital trades)
- Expand to South Gujarat (Anand, Surat belt)
- AgroSetu Finance — NBFC credit partnership
- Data intelligence product (price trends, forecasts)
- Pan-Gujarat platform — seed funding pitch

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3. Current Architecture (March 2026)

```
┌──────────────────────────────────────────────────────────────┐
│ ACTORS                                                       │
│ 👨‍💼 Broker (Father)  🛒 Wholesaler  🏭 Factory  👤 Admin      │
└──────────────────────────────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────────────────────────────┐
│ FRONTEND  (localhost:5000/public — will move to Netlify)     │
│                                                              │
│ broker-login.html     Admin login                            │
```

```
│ broker-entry.html      Trade entry                          │
│ broker-trades.html     Trade history + edit                 │
│ broker-report.html     Settlement + invoice + PDF + WA       │
│ admin.html             Master data CRUD (admin only)        │
│ price-entry.html       Daily rate entry (legacy admin)      │
│                                                             │
│ umiyaagrosetu.com ──── Netlify ──────────────────────────── │
│ index.html             Public rates page (currently static) │
│ rates.csv      ◄──── MVP 1.6: CSV-powered rates             │
└─────────────────────────────────────────────────────────────┘
                  │ fetch() with credentials
                  ▼
┌─────────────────────────────────────────────────────────────┐
│ BACKEND  Node.js + Express  (localhost:5000 → AWS EC2 later) │
│                                                             │
│ /api/broker/*      brokerRoutes.js    Trade + settlement CRUD │
│ /api/admin/*       masterDataRoutes.js  Broker/Factory/WHL/Com │
│ /api/prices/*      priceRoutes.js     Daily rate entry      │
│ /api/commodities   commodityRoutes.js  Dropdown data        │
│ /api/broadcast     broadcastRoutes.js  WhatsApp trigger     │
│                                                             │
│ middleware/brokerAuth.js   — session guard (broker routes)  │
│ middleware/adminAuth.js    — session guard (admin routes)   │
│ services/whatsappService.js — Twilio now · Meta API next    │
│ services/broadcastService.js — message formatter            │
│ db.js              — pg pool, process.env.DATABASE_URL      │
└─────────────────────────────────────────────────────────────┘
                  │ pg pool.query()
                  ▼
┌─────────────────────────────────────────────────────────────┐
│ DATABASE  PostgreSQL  (local → AWS RDS later)               │
│                                                             │
│ daily_prices       Today's factory commodity rates          │
│ price_history      Auto-logged via trigger                  │
│ commodities        Feed item catalogue                      │
│ factories          Mill/factory directory                   │
│ brokers            Broker accounts + hashed passwords       │
│ trades             All trade records (FY-aware)             │
│ brokerage_settlements  Payment records per party per FY     │
│ broker_factories   Broker ↔ factory assignments             │
│ broker_wholesalers  Broker ↔ wholesaler assignments         │
│ wholesalers        Wholesaler directory                     │
│ subscribers        WhatsApp broadcast list                  │
│ broadcasts         Message log                              │
│ system_settings    App config                               │
└─────────────────────────────────────────────────────────────┘
```

```
                    │
                    ▼
┌──────────────────────────────────────────────────────────────┐
│ EXTERNAL SERVICES                                            │
│                                                              │
│ Netlify          Static frontend hosting · SSL active        │
│ Namecheap        DNS · umiyaagrosetu.com + .in               │
│ Twilio           WhatsApp broadcast · ✅ Testing complete    │
│ Meta Cloud API   WhatsApp Business · ✅ Account active       │
│ AWS (planned)    EC2 (Node) + RDS (PostgreSQL) ap-south-1    │
│ Razorpay (future) Subscription billing Phase 2.5             │
│ TradingView      MCX Cotton + NCDEX Castor widget (free)     │
│ Google Analytics G-PCX7884MG3                                │
└──────────────────────────────────────────────────────────────┘
```

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4. Tech Stack (Current vs Planned)

Layer	Current	Planned (MVP 2.0+)
Frontend (admin/broker)	HTML/CSS/JS — localhost:5000	Same, served from AWS
Frontend (public)	Static HTML — Netlify	Static HTML — Netlify (CSV→API)
Backend	Node.js + Express — localhost	AWS EC2 — ap-south-1
Database	PostgreSQL — local	AWS RDS PostgreSQL
Auth	express-session + bcrypt	Same + JWT for Phase 2.5
WhatsApp	Twilio (tested ✅)	Meta Cloud API (account active ✅)
Billing	None	Razorpay Phase 2.5
Domain	Namecheap + Netlify	Same + `api.umiyaagrosetu.com`
CI/CD	Git push → manual	Git push → Netlify auto (frontend)
Backup	Manual	Windows Task Scheduler daily bat

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5. Open Issues & Blockers

#	Issue	Status	Action
1	LLP name approval	⏳ Pending	Wait — triggers AWS setup
2	Backend not publicly deployed	⏳ Blocked on LLP	Deploy to AWS after LLP
3	Public rates page (umiyaagrosetu.com)	🔄 MVP 1.6	Build CSV-powered page
4	Meta WhatsApp template approval	🔄 Next	Create + submit template
5	umiyaagrosetu.in not on Netlify	⏳ Pending	Add as domain alias
6	City field unpopulated	⏳ Pending	Fill in Admin → Master Data
7	Historical trade data entry	⏳ Pending	Enter from manual book records
8	₹3/bag brokerage conversation	⏳ Pending	After first invoices sent
9	Google My Business	⏳ Pending	After LLP cert received
10	Razorpay account	⏳ Pending	After LLP cert received

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6. File Structure (Current)

```
commodity-broker-backend/
├── public/
│   ├── admin.html            ← Master data CRUD (admin only)
│   ├── price-entry.html      ← Daily rate entry
│   ├── broker-login.html     ← Broker authentication
│   ├── broker-entry.html     ← Trade entry
│   ├── broker-trades.html    ← Trade history
│   └── broker-report.html    ← Settlement + invoices
├── routes/
│   ├── admin/
│   │   └── masterDataRoutes.js  ← Brokers/Factories/Wholesalers/Commodities CRUD
│   ├── brokerRoutes.js       ← All broker APIs
│   ├── priceRoutes.js        ← Daily rate APIs
│   ├── commodityRoutes.js    ← Commodity dropdown
│   ├── factoryRoutes.js      ← Factory management
│   └── broadcastRoutes.js    ← WhatsApp trigger
```

```
├── middleware/
│   ├── adminAuth.js          ← Admin session guard
│   └── brokerAuth.js         ← Broker session guard
├── services/
│   ├── whatsappService.js    ← Twilio (→ Meta API in 1.7)
│   └── broadcastService.js   ← Message formatter
├── config/
│   └── whatsappConfig.js
├── db.js                     ← pg pool
├── server.js                 ← Express app
├── .env                      ← Secrets
└── PROJECT_NOTES.md          ← Context file for Claude sessions
```

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7. Immediate Next Steps (Priority Order)

🔴 This Sprint — MVP 1.6

1. **Design `rates.csv` schema** — agree on columns before building
2. **Build CSV-powered public rates page** — Papa Parse + render factory cards
3. **Add "Export CSV" button** to `price-entry.html` — one-click download of today's rates as `rates.csv`
4. **Deploy to Netlify** — commit CSV + page → auto-deploy
5. **SEO pass** — meta tags, JSON-LD, Gujarati keywords

🟡 Parallel — WhatsApp Meta Migration (MVP 1.7)

1. **Create message template** in Meta Business Manager (Gujarati daily rates)
2. **Submit for approval** — typically 24–48 hrs
3. **Update `whatsappService.js`** — Meta Cloud API integration
4. **End-to-end test** with father's number

🟢 Waiting / Background

1. LLP certificate → AWS setup → MVP 2.0
2. Enter historical FY 2024-25 data from notebook
3. Configure `umiyaagrosetu.in` on Netlify
4. ₹3/bag conversation after first invoices sent

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8. Business Context (Unchanged)

Brokerage model: ₹2/bag from factory + ₹2/bag from wholesaler = ₹4/bag total
Settlement cycle: Annual — April to March
FY 2025-26 to date: 164 trades · 5,824 bags · ₹69,53,113 gross · ₹23,296 brokerage earned

Competitive moats:

- 30-year trust network (father knows every factory and wholesaler personally)
- Gujarati-first product — language + local knowledge barrier
- First mover in North Gujarat cattle feed digitisation
- Proven ₹2+₹2 model — zero behaviour change needed
- Data moat — first to digitise trade records owns pricing intelligence

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